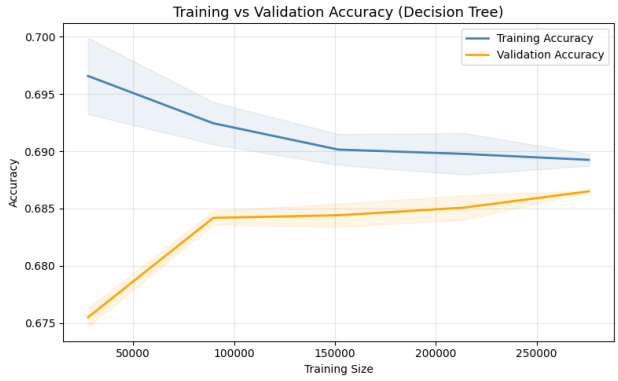
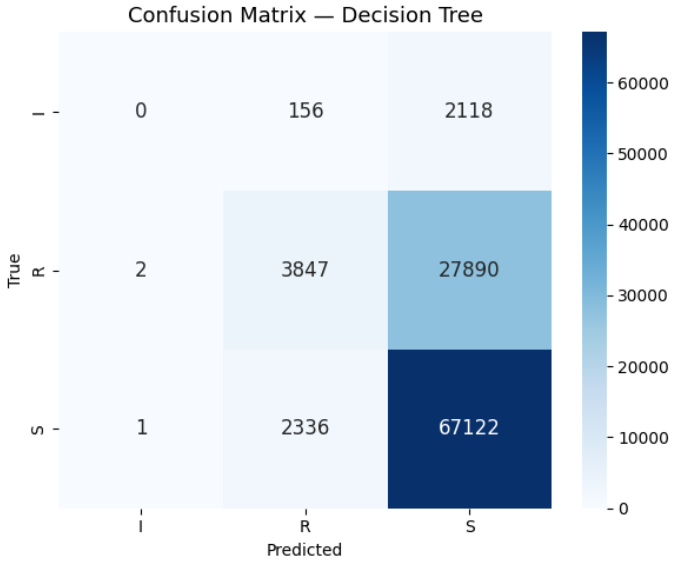
Decision Tree

Specimen + เพศ + อายุ + หอผู้ป่วย



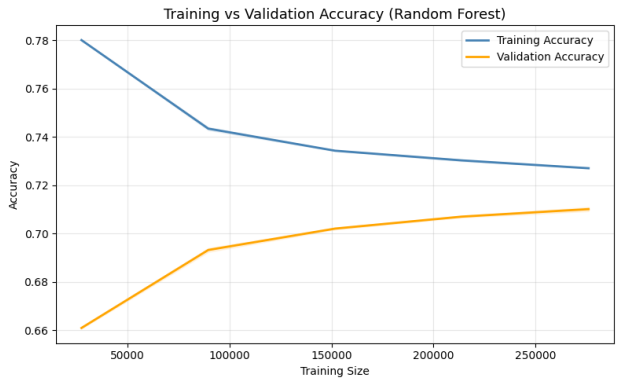


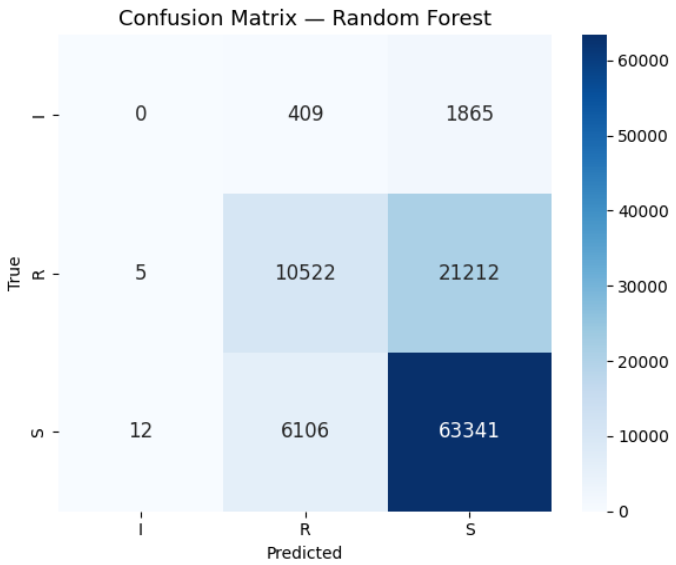
Classification Report:

Accuracy: 68.59%

	precision	recall	f1-score	support
I	0.29	0.00	0.00	2274
R	0.61	0.12	0.20	31739
S	0.69	0.97	0.81	69459
accuracy			0.70	103472
macro avg	0.43	0.36	0.34	103472
weighted avg	0.65	0.69	0.60	103472

Random Forest



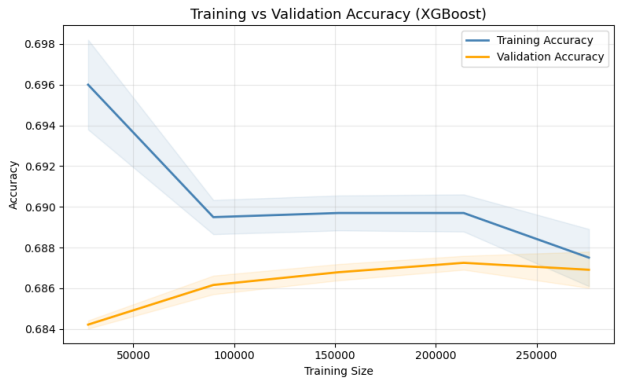


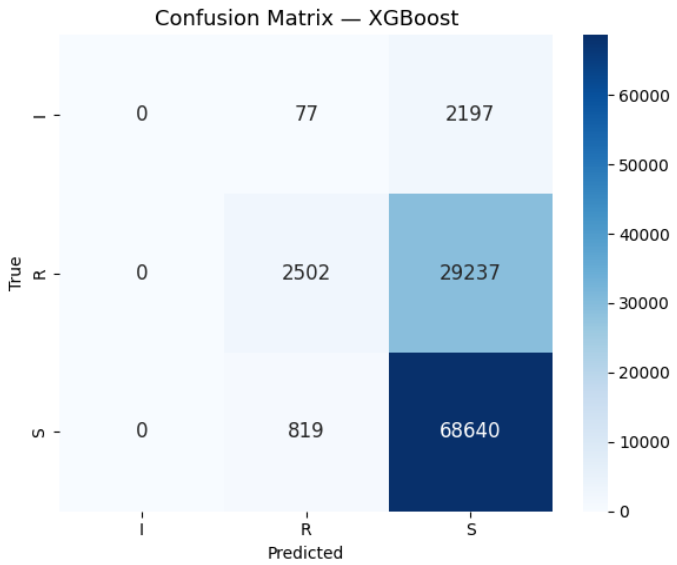
Classification Report:

Accuracy: 71.38%

	precision	recall	f1-score	support
I	0.00	0.00	0.00	2274
R	0.62	0.33	0.43	31739
S	0.73	0.91	0.81	69459
accuracy			0.71	103472
macro avg	0.45	0.41	0.41	103472
weighted avg	0.68	0.71	0.68	103472

XGBoost



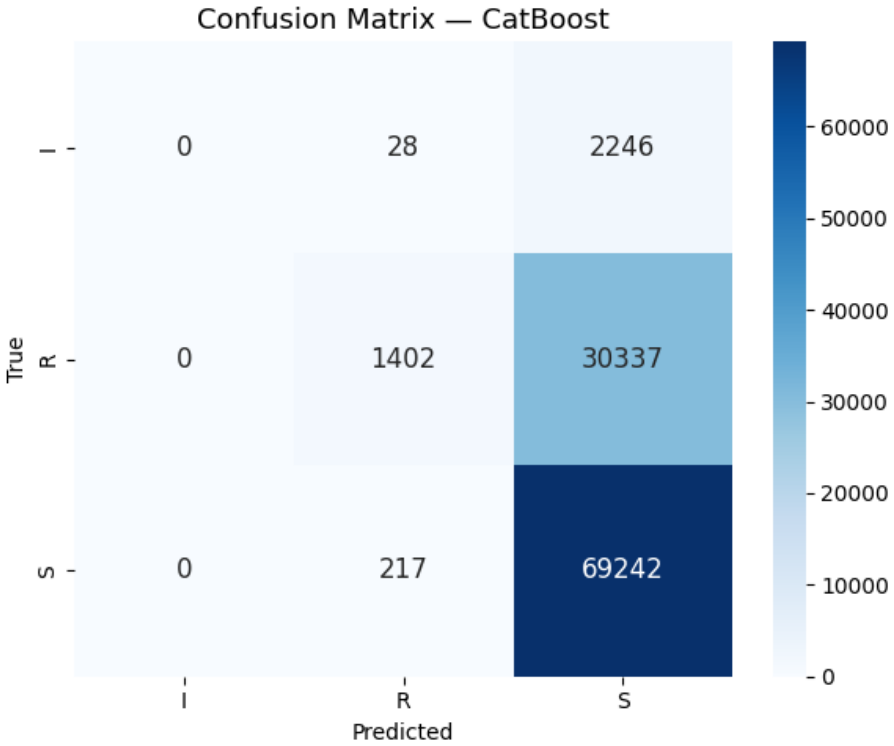


Classification Report:

Accuracy: 68.75%

	precision	recall	f1-score	support
I	0.00	0.00	0.00	2274
R	0.74	0.08	0.14	31739
S	0.69	0.99	0.81	69459
accuracy			0.69	103472
macro avg	0.47	0.36	0.32	103472
weighted avg	0.69	0.69	0.59	103472

CatBoost



Classification Report:

Accuracy: 68.27%

	precision	recall	f1-score	support
I	0.00	0.00	0.00	2274
R	0.85	0.04	0.08	31739
S	0.68	1.00	0.81	69459
accuracy			0.68	103472
macro avg	0.51	0.35	0.30	103472
weighted avg	0.72	0.68	0.57	103472

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STEP 6.7 — Model Comparison

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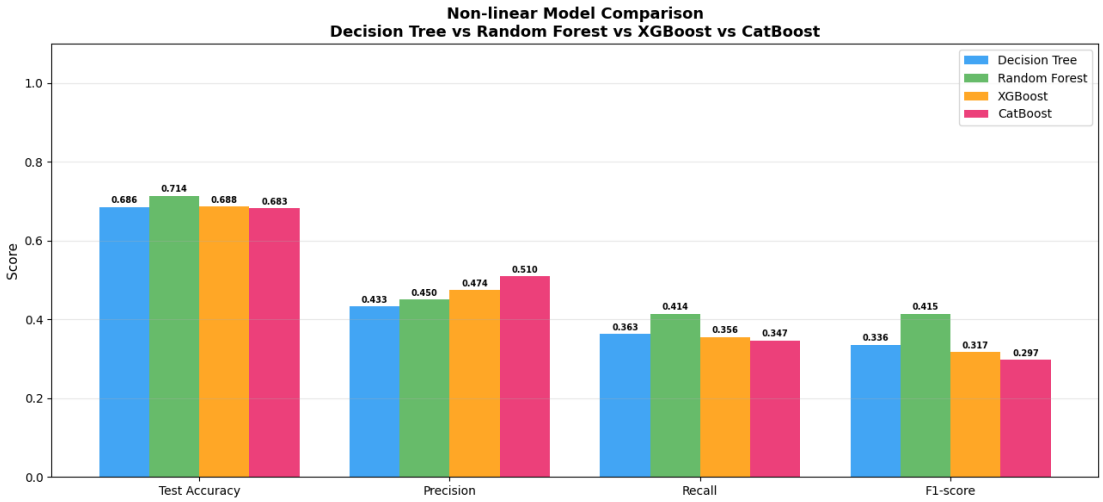
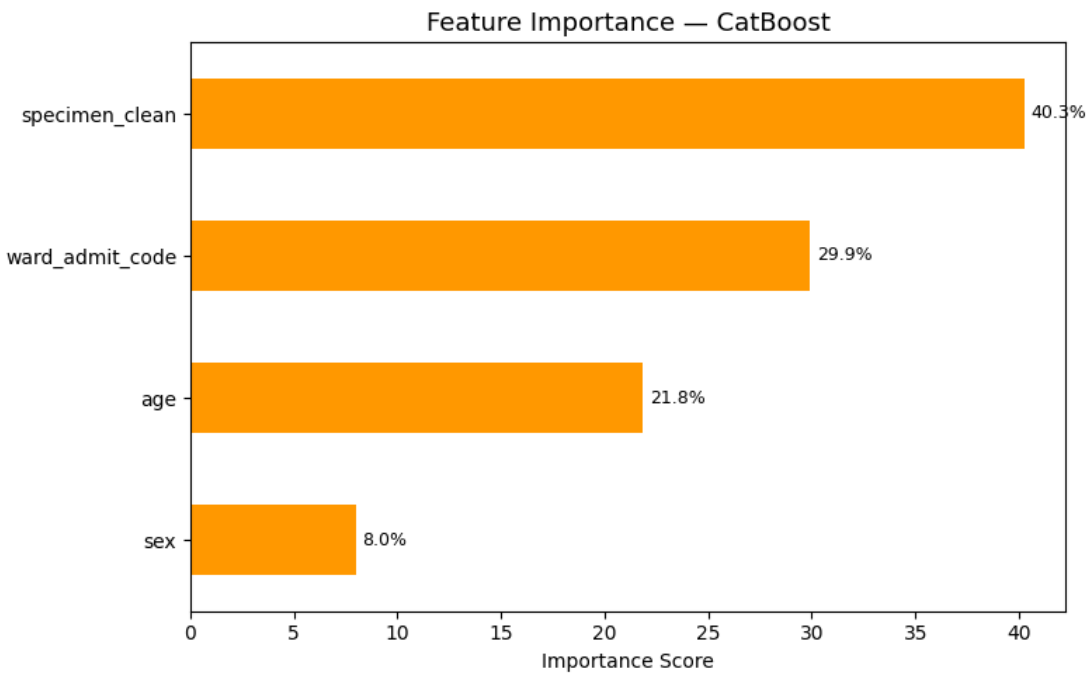
Model	Test Acc	Precision	Recall	F1
Decision Tree	0.6859	0.4326	0.3625	0.3360
Random Forest	0.7138	0.4502	0.4145	0.4147 ✅
XGBoost	0.6875	0.4741	0.3557	0.3174
CatBoost	0.6827	0.5104	0.3470	0.2975





Best Model : Random Forest

Test Accuracy : 71.38%

Macro F1 : 0.4147

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